

Yue Cao

Data Analysis, Business Analysis

📍 Canberra

☎️ +86 15209860534

@ 15209860534@163.com

Skills

Python

Pandas, numpy, machine learning, vectorbt

Data Analysis

Statistical and mathematical foundations, Data processing and cleaning, Data visualization, Business insights, Strategic analysis

SQL

Data querying, Database design, Big data processing

Photoshop

Chart creation, Poster design, Layout design

Certifications

Contracted Photographer, Visual China 2020

Outstanding Member of the Student Union 2018

Second Prize, Hu Bing Cup New Media Creative Product Design Competition 2019

Third Prize, Anhui Province 3rd Campus Media Elite Competition for Outstanding New Media in Universities 2019

Second Prize, 4th University Student Network Culture Festival, Hefei University of Technology 2019

Second Prize, Anhui Province Service Outsourcing Innovation and Entrepreneurship Competition 2019

Outstanding League Member of the Youth League Branch 2020

Awards

Outstanding Member of the New Media Center, University Youth League Committee, Hefei University of Technology 2018

Second Prize, Hu Bing Cup New Media Creative Product Design Competition 2019

Third Prize, Anhui Province 3rd Campus Media Elite Competition for Outstanding New Media in Universities 2019

Second Prize, 4th University Student Network Culture Festival, Hefei University of Technology 2019

Second Prize, Anhui Province Service Outsourcing Innovation and Entrepreneurship Competition 2019

Outstanding League Member of the Youth League Branch 2020

Interests

Photography

Go (Weiqi)

Video Editing

Education

Hefei University of Technology

Materials Science and Engineering - Metal Materials Engineering

2018.09-2022.06

Bachelor's

Australian National University

Business Information System

2024.02-2025.11

Master's

With an interdisciplinary background combining a Bachelor's degree in **Engineering** and a Master's degree in **Business**, I possess both technical expertise and strategic business acumen. My engineering education provided me with a solid mathematical foundation and programming skills, making me adept at building machine learning models with **Python**, managing databases efficiently with **SQL**, and transforming complex data into intuitive visualizations using tools like **Matplotlib** and **Tableau**. My business training has sharpened my *market insight* and *strategic analysis capabilities*, enabling me to extract business value from data to drive decision-making.

Experience

ANU Advancement

Data Analysis Internship

Jul 2025 - Oct 2025

Canberra

Literature Review & Methodology Development:

Conducted a systematic literature review of academic and industry reports on philanthropic fundraising to research advanced statistical modeling methods (e.g., Survival Analysis) and machine learning algorithms (e.g., Gradient Boosting) for donor prediction, establishing a data-driven technical approach for the project.

Data Processing & Exploratory Data Analysis (EDA):

Utilized Python (Pandas, NumPy) to perform data cleaning, integration, and feature engineering on over 100,000 records from the alumni Customer Relationship Management (CRM) system. Identified key predictive factors significantly correlated with donation potential—such as historical donation frequency, event participation, and geographic distribution—through statistical testing and visualization (Matplotlib, Seaborn).

Machine Learning Modeling & Validation:

Led the design and implementation of machine learning classification models (Logistic Regression, Random Forest) using Scikit-learn to predict the likelihood of high-net-worth donors. Achieved a final model AUC of 0.85 through cross-validation and grid search for hyperparameter tuning, significantly improving the accuracy of identifying the target audience.

Data Visualization & Dashboard Development:

Designed and built a dynamic dashboard in Tableau to visualize model predictions, donor profiles, and key performance indicators (KPIs). The automated reports provided the fundraising team with intuitive data insights, effectively supporting strategic decision-making.

Deliverables & Strategic Recommendations:

Authored and presented a detailed technical report containing data insights, model architecture, performance evaluation, and deployment recommendations. The proposed model integration and data dashboard are projected to increase the efficiency of screening potential donors by approximately 30%.

Projects

Spam SMS Classification and Prediction Project (Based on XGBoost)

Developed a machine learning-based SMS classification system

Dec 2024

Tech Stack:

Python, pandas, numpy, scikit-learn, XGBoost, natural language processing (NLTK), data visualization (matplotlib, seaborn, WordCloud)

Project Details:

Conducted exploratory analysis and visualization of a dataset with over 5,500 SMS entries.

Designed and implemented a complete text preprocessing pipeline, including cleaning, normalization, stop-word removal, and stemming.

Applied TF-IDF vectorization for feature engineering to extract key text features.

Built and optimized an XGBoost classification model for accurate spam SMS detection.

Achievements:

Achieved a classification accuracy of 97.19% on the test set.

Attained an F1 score of 0.98 for normal SMS and 0.88 for spam SMS identification.

Developed an efficient text processing workflow, significantly improving model performance.

Visualized differences in text features between spam and normal SMS using word clouds.

Quantitative Trading Strategy Development and Live Trading

RSI + Bollinger Bands Trading Strategy

Nov 2024 - Jan 2025

Tech Stack:

Python, pandas, numpy, talib, vectorbt, ccxt, plotly

Project Details:

Combined **Relative Strength Index (RSI)** and **Bollinger Bands** to identify overbought and oversold market conditions.

Used the **vectorbt** library for indicator calculations.

Conducted backtesting with historical data to evaluate strategy performance, including annualized returns, maximum drawdown, and win rate, and visualized account equity curves.

Adjusted RSI thresholds to optimize the precision of trading signals.

Achievements:

Executed live trading via **freqtrade** connected to an exchange.

Monitored simulated trading performance, optimized stop-loss/take-profit mechanisms, and reduced risks from market volatility.

Achieved a monthly return of 2% and an annualized return of 25% in backtesting trading.

Strategic Report

Leveraging Analytics & AI in Retail (Woolworths Case Study)

Jun 2025

Authored a strategic analysis for Woolworths Group, proposing AI-driven solutions for key retail challenges (inventory, customer churn, marketing ROI, omnichannel integration) and outlining a phased implementation roadmap guided by the Analytics Maturity Model.

Key Recommendations & Proposed Solutions:

Inventory Optimization:

Applied Machine Learning for demand forecasting (aiming for 15-30% accuracy uplift) and Prescriptive Analytics for efficient stock management.

Customer Retention:

Utilized RFM analysis & clustering for customer segmentation, and NLP on customer feedback to identify churn drivers and improve loyalty.

Marketing Effectiveness:

Advocated for advanced Multi-Touch Attribution models and A/B testing to enhance campaign performance and ROI.

Omnichannel Integration:

Proposed Unified Commerce Platforms, IoT for in-store analytics, and Generative AI for creating seamless, personalized customer experiences.

Strategic Roadmap:

Developed a 3-phase implementation plan emphasizing robust data governance, scalable cloud infrastructure, workforce upskilling, ethical AI deployment, and defined KPIs for measuring impact.

Core Skills Demonstrated:

Strategic Analysis, Business & Retail Analytics, AI/ML Solution Design, Data-Driven Strategy, Omnichannel Development, Implementation Planning.

Campus Experience

Hefei University of Technology (HFUT) New Media Center

Product Operations Director

Jul 2019 - Apr 2020

Key Achievements:

Drove platform growth through refined operational strategies, resulting in an **increase of over 2,000 followers on QQ/Weibo and over 5,000 followers on WeChat**.

Data-Driven Operations:

Leveraged **Excel** for in-depth analysis of user behavior data to build accurate user personas and optimize content strategy, which **significantly boosted user interaction rates and platform activity**.

Operational Framework Development:

Introduced a **KPI performance evaluation system** to quantitatively track interaction data (e.g., views, click-through rates, shares) for various content formats, using data feedback to continually iterate and **enhance the platform's overall influence**.

Process Optimization & Efficiency Improvement:

Established and optimized standard operating procedures, clarifying responsibilities at each stage and **improving team collaboration efficiency and content quality**.